

[Skip to main content](#)

- [Welcome](#)
- [Sample Content](#)
 - [Meteo 101](#)
 - [Meteo 241](#)
 - [Meteo 361](#)
 - [Meteo 410](#)
- [Our Alumni](#)
- [Join Us!](#)

Learning Weather by the Penn State Meteorology Department

Search form

Search

Sample Content

- [Meteo 101: Sample Content](#)
- [Meteo 241: Sample Content](#)
- [Meteo 361: Sample Content](#)
 - [Lesson 1. Meeting the Mesoscale](#)
 - [More about Spatial Scales](#)
 - [Subdividing the Mesoscale](#)
 - [Time Scales Versus Durations](#)
 - [The Rapid Refresh Model](#)
 - [The 3-kilometer NAM](#)
 - [Case Study: May 10, 2010](#)
 - [Lesson 3. Sizing up the Synoptic Scale](#)
- [Meteo 410: Sample Content](#)

For Internal Use

- [login](#)

More about Spatial Scales

This is a sample lesson page from the [Certificate of Achievement in Weather Forecasting](#) offered by the Penn State Department of Meteorology. Any questions about this program can be directed to: [Steve](#)

Prioritize...

When you've completed this page, you should be able to 1) distinguish between planetary scale, synoptic scale, mesoscale, and microscale features based on their size definitions, 2) identify some common features in each size scale, and 3) place features on weather maps into the proper size scale using reference measurements.

Read...

The spatial scales of weather systems run the gamut from planetary scale to microscale. Before we get into defining each specific scale, I should point out that none of them have universally accepted definitions. That's right, the "boundaries" of each size scale can be somewhat murky. Therefore, think of the size scales more as a continuum, instead of having hard, fixed boundaries. In any event, I still want to give you some general guidelines, and in this course, we'll base our definitions on some of the more commonly used criteria. Just keep in mind that the exact boundaries are somewhat artificial.

The **planetary scale** typically includes long waves, which have wavelengths exceeding 5000 kilometers (about 3000 miles). For example, the analysis of the daily average 500-mb heights on May 10, 2010 (see below), reveals several long waves encircling the Northern Hemisphere. Note the long-wave trough over eastern North America and the long-wave ridge farther downstream over the Atlantic Ocean. Technically speaking, the wavelength of this trough-ridge couplet is the distance between the trough axis over eastern North America and the trough axis off the west coast of Europe (marked by the dashed white lines). This distance is right around 5000 kilometers, so it falls into the planetary scale.

 Daily average 500-mb heights over the Northern Hemisphere on May 10, 2010

The average daily 500-mb heights on May 10, 2010. Note the long-wave trough and ridge over eastern North America and the North-central Atlantic Ocean, respectively. This long wave qualifies as a feature on the planetary scale.

Credit: Earth System Research Laboratory

Next in our spectrum of spatial scales is the **synoptic scale**, which refers to features ranging from about 1000 kilometers (about 600 miles) to 5000 kilometers. However, I want to again emphasize some murkiness here. Many meteorologists take the smaller end of the synoptic-scale to be 2000 kilometers (about 1200 miles), so just realize that when you encounter features between 1000 kilometers and 2000 kilometers, you may find some disagreement about their classifications. Regardless of that murkiness, you should already be familiar with many synoptic-scale features. The mid-latitude high- and low-pressure systems that you've studied in previous courses, along with warm and cold fronts associated with mid-latitude cyclones are typically considered synoptic scale features, when measured by their lengths.

That qualifier I added at the end, "when measured by their lengths" is very important because whenever you're attempting to categorize the scale of weather systems, always keep in mind that your classification depends on the axis along which you're measuring. For example, if we look at the [surface analysis from 03Z on August 23, 2015](#), the length cold front that snakes from the Upper Midwest back through the Rockies qualifies as synoptic scale (and that's typical of most cold fronts). However, [cross-sectional views of fronts](#) associated with mid-latitude cyclones reveal that the air motions along (and near) the front are much smaller, and are typically less than 1000 kilometers, so they're smaller than synoptic scale.

The bottom line is that any classification of the spatial scale of weather systems often depends on the horizontal axis along which you focus your analysis. You may find that along its major (longer) axis, a feature fits into one size scale, but along its minor (shorter) axis, a feature fits into another size scale. It's fairly common

for surface fronts to be synoptic-scale in terms of their lengths (major axis), but have vertical motions that occur across the front (minor axis) which qualify as mesoscale.

Speaking of the mesoscale, it's time to finally complete our definition. **Mesoscale** weather features are between roughly 2 kilometers (1.2 miles) and 1000 kilometers. Many various mesoscale weather features exist, and we'll study a lot of them in this course. For now, however, we'll use a thunderstorm as a common example of a mesoscale weather feature. As you'll soon see, meteorologists actually subdivide the mesoscale even further, and we'll get into more details on that in the next section.

Finally, **microscale** weather features are those that span less than two kilometers. Even though we'll study tornadoes in depth in this course, technically, most of them are microscale features. Very few tornadoes exceed the two kilometers in width needed to qualify them as mesoscale features.

That's a quick run down on spatial scales from planetary scale to microscale. Up next, we'll take a closer look at how meteorologists subdivide the mesoscale. Before you move on, however, an important skill that you need to develop is the ability to identify features on weather maps, and classify their size scale properly. Check out the *Key Skill* section below for some important discussion and tips about properly sizing things up.

Key Skill...

What's the best way to classify the size scale of various weather features on weather maps? If the map happens to have a distance scale, it's straightforward -- just use the distance scale to estimate the size of the feature. But, in reality most weather maps and model graphics don't contain distance scales. So, what can you do to easily estimate the size of a weather feature?

Perhaps the simplest way is to use a **reference measurement**, which is a method of measurement that compares an object of known length with the object you're measuring. For example, the distance across the United States (west to east) across the northern portion of the country (including New England) is a bit less than 5000 kilometers. For simplicity, let's call it 5000 kilometers exactly. So, if an object is larger than the distance across the United States, it's larger than 5000 kilometers, meaning it's a planetary-scale feature.



The west-east distance across the northern United States (including New England) is a bit less than 5000 kilometers, but for practical purposes we can use this as a reference measurement to determine whether features are large enough to be considered planetary scale.

Credit: Google

What are some other reference measurements that can help us classify weather features?

- The west-east distance across Pennsylvania is approximately 500 kilometers
- The west-east distance across Utah (the wide part) is approximately 500 kilometers
- The north-south distance across Kansas is approximately 300 kilometers
- The west-east distance across central Vermont is approximately 100 kilometers

How can we use these references in practice? If, for example, a feature is more than "two Pennsylvania's" or "two Utah's" in size, then it's more than 1000 kilometers, and is a synoptic-scale feature. If it's smaller than that, it's a mesoscale feature (or microscale, but it would be hard to identify microscale features on maps showing the entire United States).

For example, check out the [300-mb analysis from 12Z on September 8, 2015](#), and note the jet streak over western Canada. What size scale does this feature fit into? If we use our nearest reference measurement, we can tell that the jet streak is more than "two Utahs" long, so it's more than 1000 kilometers long. It's also obviously smaller than the west-east distance across the United States (around 5000 kilometers), so it must be a synoptic-scale feature. Now, what if we wanted to classify only the core of that jet streak (the white area of fastest wind

speeds near its center)? The core looks to be less than "one Utah" long, so it's less than 500 kilometers -- certainly, a mesoscale feature.

Obviously, this process requires some visual estimation, and is not exact, but it's a quick and useful way to "size up" a weather feature. If you're worried about being inexact in a borderline case, don't be. Remember that the boundaries between scales are somewhat murky anyway. Hopefully the handful of reference measurement examples listed above give you some tools that you can use for features around the United States.

[< Lesson 1. Meeting the Mesoscale Subdividing the Mesoscale >](#)



Adapted by David Babb (Assistant Professor, Department of Meteorology, College of Earth and Mineral Sciences

© 2019 [The Pennsylvania State University](#), from original course materials by Lee M. Greci.

This courseware module is part of Penn State's College of Earth and Mineral Sciences' [OER Initiative](#).

Except where otherwise noted, content on this site is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported License](#).

The College of Earth and Mineral Sciences is committed to making its websites accessible to all users, and welcomes comments or suggestions on access improvements. Please send comments or suggestions on accessibility to the [site editor](#). The site editor may also be contacted with questions or comments about this course.



The John A. Dutton e-Education Institute is the learning design unit of the [College of Earth and Mineral Sciences](#) at [The Pennsylvania State University](#).

Navigation

- [Home](#)
- [News](#)
- [About](#)
- [Contact Us](#)
- [Programs and Courses](#)
- [People](#)
- [Resources](#)
- [Services](#)
- [Login](#)

EMS

- [College of Earth and Mineral Sciences](#)
- [Department of Energy and Mineral Engineering](#)

- [Department of Geography](#)
- [Department of Geosciences](#)
- [Department of Materials Science and Engineering](#)
- [Department of Meteorology](#)
- [Earth and Environmental Systems Institute](#)
- [Energy Institute](#)
- [Institute for National Gas Research](#)

Programs

- [Online Geospatial Education Programs](#)
- [iMPS in Renewable Energy and Sustainability Policy Program Office](#)
- [BA in Energy and Sustainability Policy Program Office](#)
- [M.Ed. in Earth Sciences Program Office](#)

Related Links

- [Dutton Community Yammer Group](#)
- [Penn State e-Learning Cooperative](#)
- [Penn State World Campus](#)
- [Web Learning @ Penn State](#)



[2217 Earth and Engineering Sciences Building, University Park, Pennsylvania 16802](#)
800-555-1212 | [Contact Us](#)

[Privacy & Legal Statements](#) | [Copyright Information](#)

The Pennsylvania State University © 2014